Mapping intrinsic laryngeal muscle activity using HD-sEMG

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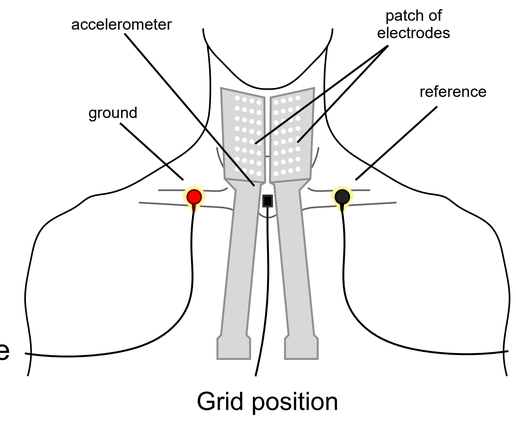
Introduction. Intrinsic laryngeal muscles (ILMs) regulate vocal folds characteristics in human phonation. Among these, the cricothyroid (CT) muscle contribute in controlling voice fundamental frequency (F0) (Gay et al., 1972).

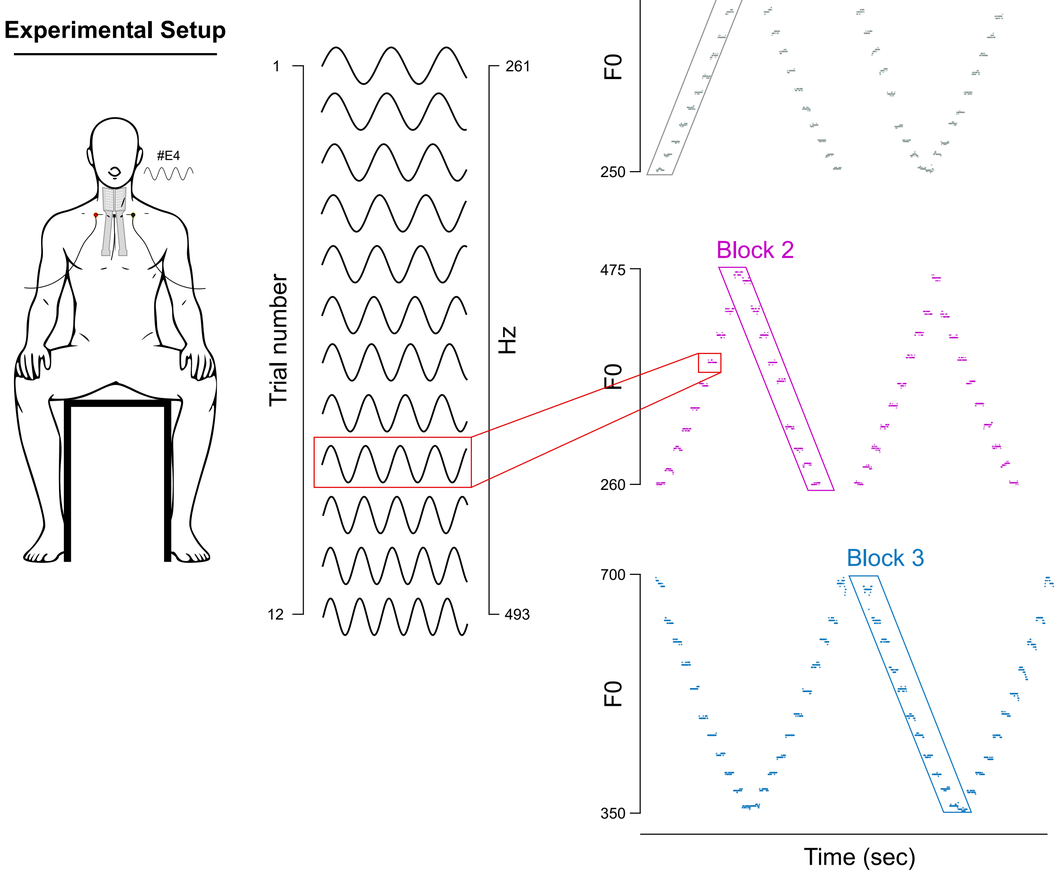
High-density surface EMG (HD-sEMG) instead of the traditional needle EMG, offers a promising method for revealing task- and pitch-dependent activation patterns of laryngeal muscles (Martinez et al., 2026).

Aim. To explore CT spatial activation patterns during pitch modulation.

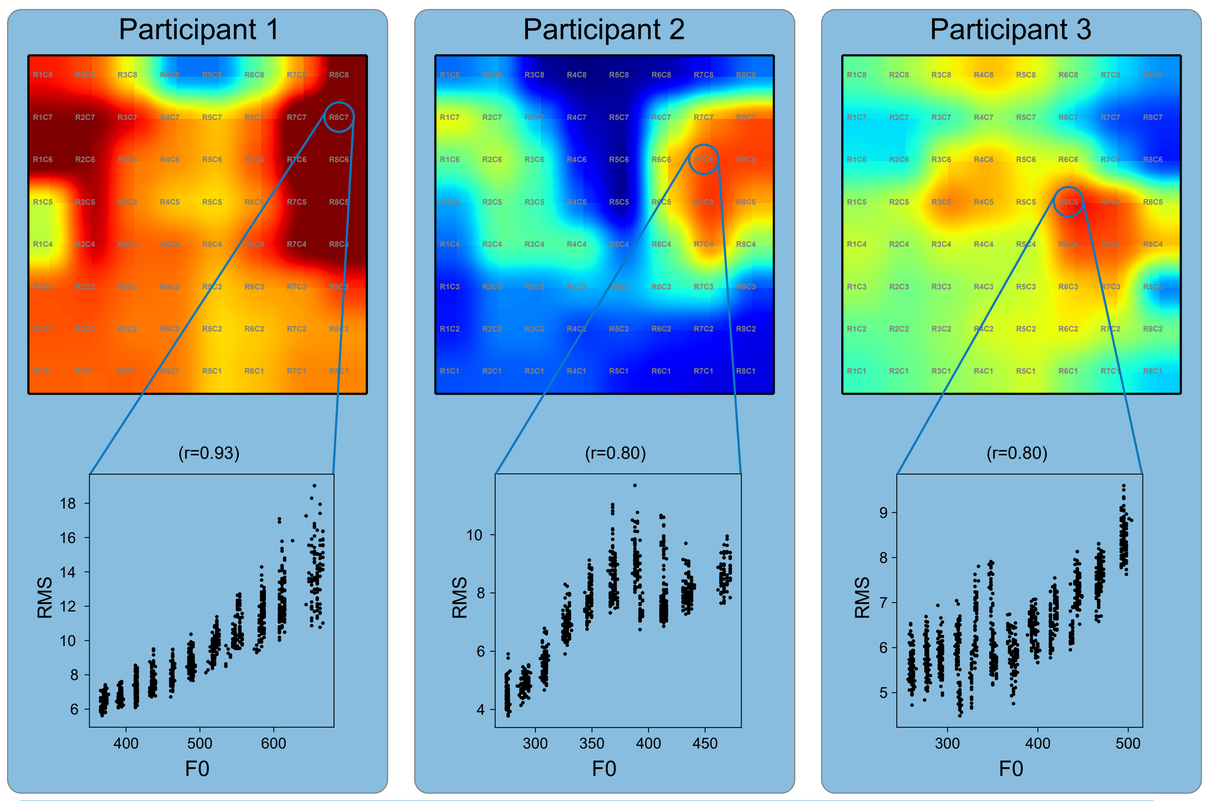
Methods. Task: 7sec of sustained /a/ phonation across 12 semitones within participants' vocal range. The task was recorded on female participants (n=3) with a 64-channel HDsEMG (TMSi Artinis) with sfreq = 4096 Hz. Larynx apparatus vibratory activity were acquired through the use of an accelerometer.

Analysis. Average reference were selected during recordings. Raw data were bandpass filtered (20-700Hz) and 500ms segments with 250ms overlap were created for later analysis. F0 was estimated through accelerometer data. Correlation analysis and Root Mean Squared (RMS) were performed on the segments.





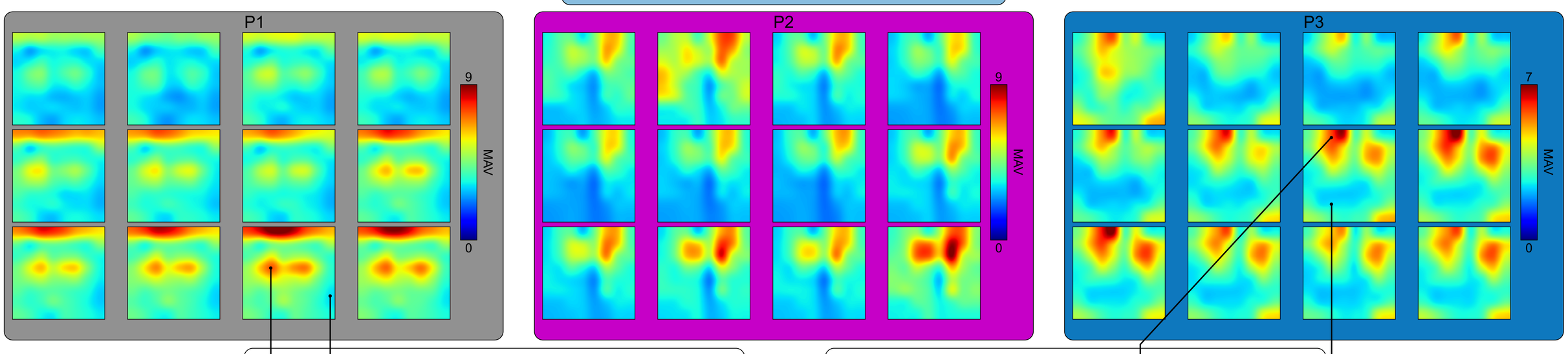
Correlation between EMG activity and F0

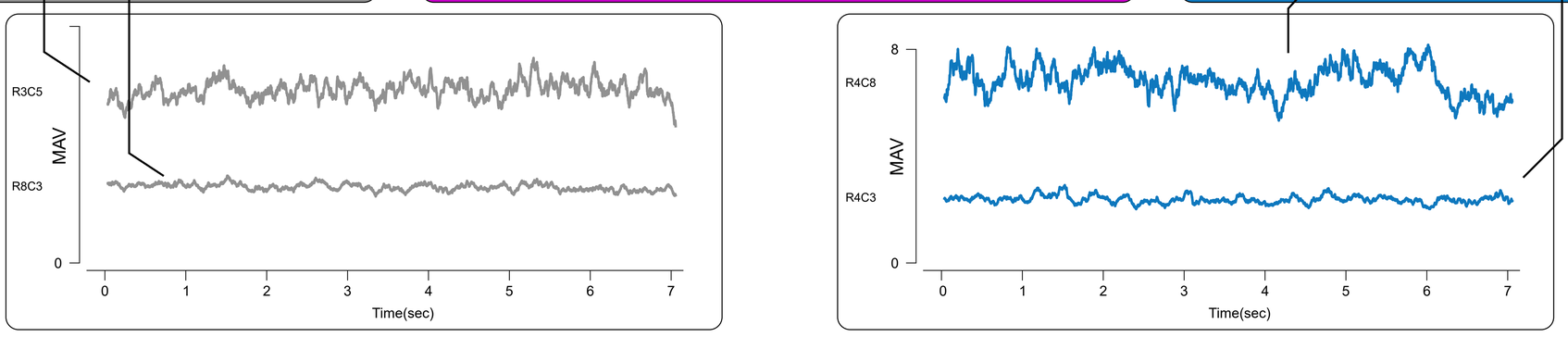


Observations. The electrodes with the highest correlation were not those that show the stronger EMG contraction.

Results I. The higher F0, the higher the EMG amplitude by channel.

Average muscle contraction by pitch





Results II. The observable pattern of activity is an increase in the RMS of the EMG in the channels targeting the cricothyroid muscle. Despite being slightly different among participants, it suggests the possibility of isolating the activity of the two muscles.

Take home message. This preliminary results, together with the innovative experimental setup, point towards the feasibility of using HD-sEMG to explore intrinsic laryngeal muscle dynamics

References

Gay, T., Strome, M., Hirose, H., & Sawashima, M. (1972). Electromyography of the intrinsic laryngeal muscles during phonation. Annals of Otology, Rhinology & Laryngology, 81(3), 401-409.

Martínez, J., Ibarra, E., & Zañartu, M. (2026, April). Estimating Cricothyroid Muscle Activation from HDsEMG Using a Neural Network–Based Framework: A Case Study. In 2026 IEEE Medical Measurements & Applications (MeMeA) (pp. 1-6). IEEE.

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